

→ Escape Sequence

Escape sequences are combinations of characters that are used to represent special characters in strings.

Code	Result
\'	Single Quote
\n	New Line
\t	Tab

```
x = "hello \"haiii \" welcome"
print(x)
```

```
hello "haiii " welcome
```

→ OPERATORS

1. Arithmetic operators

Operators	Meaning	Example	Result
+	Addition	$4 + 2$	6
-	Subtraction	$4 - 2$	2
*	Multiplication	$4 * 2$	8
/	Division	$4 / 2$	2
%	Modulus operator to get remainder in integer division	$5 \% 2$	1
**	Exponent	$5 ** 2 = 5^2$	25
//	Integer Division/ Floor Division	$5 // 2$	2

2. Comparison operators

Operators	Meaning	Example	Result
<	Less than	$5 < 2$	False
>	Greater than	$5 > 2$	True
<=	Less than or equal to	$5 <= 2$	False
>=	Greater than or equal to	$5 >= 2$	True
==	Equal to	$5 == 2$	False
!=	Not equal to	$5 != 2$	True

3. logical operators

Operator	Meaning	Example	Result
and	Logical and	$(5 < 2) \text{ and } (5 > 3)$	False
or	Logical or	$(5 < 2) \text{ or } (5 > 3)$	True
not	Logical not	not $(5 < 2)$	True

- and gate: (Multiplication)

2 input AND Gate

A	B	A.B
0	0	0
0	1	0
1	0	0
1	1	1

- Or gate (Addition)

2 input OR gate

A	B	A+B
0	0	0
0	1	1
1	0	1
1	1	1

- And gate

2 input NOT gate

A	$\overline{A}$
0	1
1	0

- Membership operator: in , not in
- Identity operator : is , is not

→ Input

`input()` function is used to take user input from the keyboard during the execution of a program. It allows the program to interact with the user by accepting data entered through the console.

1. `eval()`
2. `int()`
3. `float()`
4. `str()`

```
x=int(input("enter a number"))
```

enter a number

```
x=float(input("enter a number"))
```

enter a number

```
x=str(input("enter your name"))  
y=str(input("enter your pet name"))
```

enter your name

enter your pet name

```
x=eval(input("enter your name"))
```

enter your name

```
x=eval(input("enter a number"))
```

enter a number